

From Generation to Embedding: A Transformer Study of Edit Distance and Error Correction Codes

Hsin Ya Lin

Institute of Communications, National Yang Ming Chiao Tung University

Abstract

This report studies how well Transformer models handle error correction code sequences, measured against traditional algorithms such as the BCJR decoder. The work moved through three architectures. A generative sequence-to-sequence Transformer produced full bit-level metrics and tracked edit distance closely, with a Pearson correlation of 0.8589, yet its Block Error Rate of 0.9999 made it useless as a practical decoder. Removing the decoder and training an encoder-only Siamese network with a Triplet Margin Loss fixed the instability and reached the lowest absolute error of any learned model tested, an RMSE of 2.092 and a Triplet Accuracy of 0.7238, at the cost of no longer emitting sequences. A third attempt added a Connectionist Temporal Classification head to recover bit sequences again; it did not converge, and the reasons are documented as future work. The central result is that attention can map discrete sequence mutations into a measurable geometric space, but the choice between predicting a sequence and embedding it carries a sharp tradeoff in which metrics become available.

Problem and Metrics

Edit distance is the smallest number of insertions, deletions, and substitutions that turn one sequence into another. Error correction asks a related question: given a corrupted codeword, recover the original. Two families of metrics matter here. Bit Error Rate and Block Error Rate count how many bits, or whole blocks, come out wrong after decoding. Root Mean Square Error, Mean Absolute Error, and the Pearson and Spearman correlations measure how well a model's predicted distance tracks the true Levenshtein distance. A model that only outputs a number can be scored

on the second family but not the first. This distinction drove every design decision that follows.

Phase One: Generative Sequence-to-Sequence Baseline

The first model was a standard sequence-to-sequence Transformer with an autoregressive decoder, trained with a positional binary cross-entropy loss. It worked well enough to score across both metric families. On the held-out set it reached an RMSE of 2.4817, an MAE of 1.9312, a Pearson correlation of 0.8589, and a Spearman correlation of 0.8557. Because it emits actual bits, it also produced a Bit Error Rate of 0.3716 and a Hamming Loss of 0.3716.

Read the right way, the generated sequences were close to correct. When each output is compared to its target by edit distance, the predictions follow the diagonal across the full range of severities, which is the 0.8589 Pearson figure in visible form. Figure 1 shows this fit. The encoder representations carry the same structure: a Principal Component Analysis of the encoder states, in Figure 2, captures 86.3 percent of the variance in two components and arranges sequences by edit distance rather than at random.

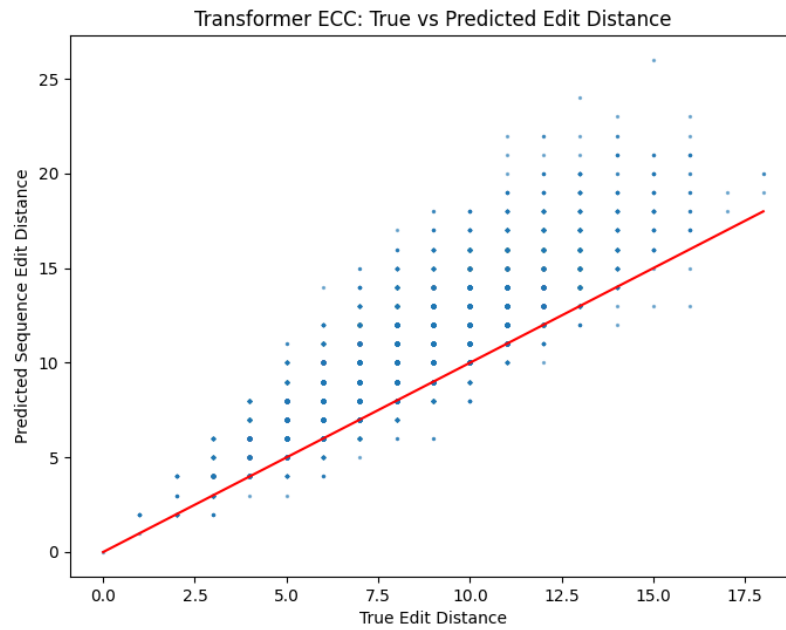


Figure 1: True edit distance against the edit distance of the generated sequence. The predictions track the diagonal, which is the source of the strong rank correlation.

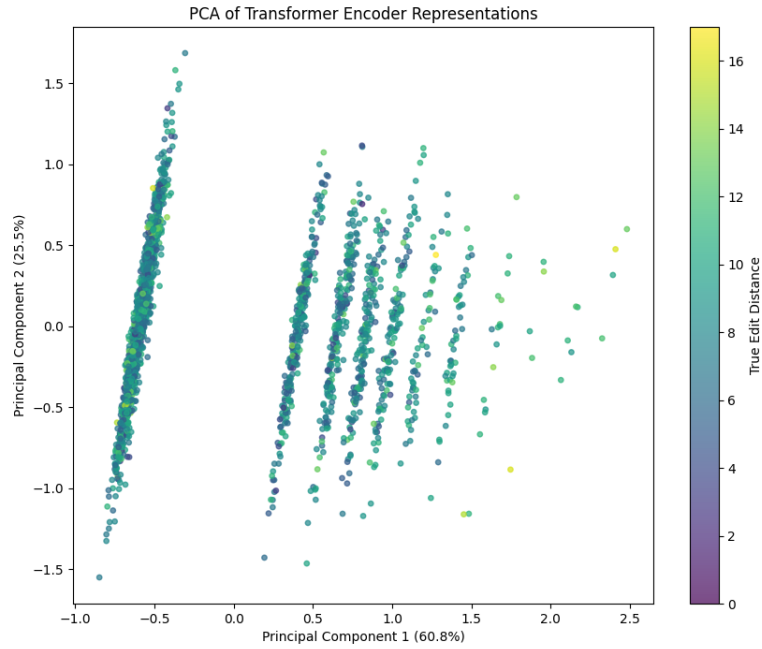


Figure 2: Principal Component Analysis of the generative encoder states, colored by true edit distance. Two components hold 86.3 percent of the variance.

The problem sat in one number. The Block Error Rate was 0.9999, meaning almost no block came out fully correct. The cause is structural rather than a tuning failure. Error correction involves insertions and deletions, and a single uncorrected deletion shifts every later bit one position to the left. A positional loss reads that shift as a wholesale failure and punishes a sequence that is, in edit terms, nearly right. The decoder never escaped this penalty, so its bit-level output stayed unusable even while its distance estimates looked reasonable. Figure 3 makes the gap plain: the bit rate sits near 0.37 while the block rate stands at almost one.

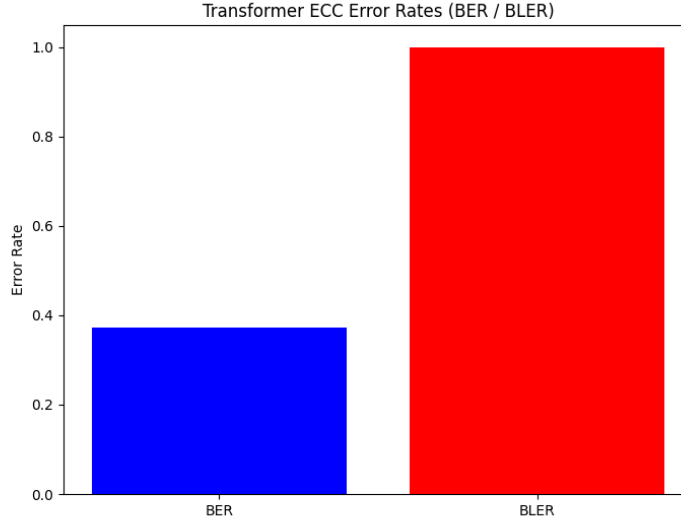


Figure 3: Bit Error Rate and Block Error Rate for the generative Transformer. The block rate near unity reflects shift-induced cascading errors rather than isolated bit mistakes.

One variant replaced sequence generation with a head that regresses the edit distance directly as a single number. It did not hold up. The predictions saturated near five operations and ignored larger distances, which dragged the correlation down to 0.311 and the RMSE up to 3.772. That collapse is why the fair reading of this phase comes from the generated sequences rather than a regressed scalar, and the numbers reported here follow that reading.

Phase Two: Siamese Transformer Network

The shift penalty is a property of generation, not of attention, so the second design stopped generating. The decoder was removed, and the encoder was reused as a Siamese network that maps each sequence into a 256-dimensional Euclidean space. Two sequences are compared by the distance between their vectors, which sidesteps the alignment problem entirely.

The encoder uses an embedding dimension of $d_{model} = 256$, eight attention heads, and four layers. Mean pooling reduces the per-position encoder output to a single vector per sequence.

Training ran for 30 epochs with the AdamW optimizer. The objective was a Triplet Margin Loss over an anchor, a correlated positive, and a corrupted negative:

$$\mathcal{L}(a, p, n) = \max\{d(a, p) - d(a, n) + \text{margin}, 0\}, \quad (1)$$

where d is the Euclidean distance and the margin is 1.0. To report RMSE and MAE on the edit distance scale, a one-dimensional linear regression maps the continuous embedding distance onto the discrete Levenshtein distance. This is a calibration step; the network itself learns geometry, not a number.

Across 10,000 triplets the model placed correlated pairs at an average distance of 2.7128 and corrupted pairs at 3.3750. The resulting Triplet Accuracy, the fraction of triplets where the positive is closer than the negative, was 0.7238. Figure 4 shows the two distance distributions pulling apart.

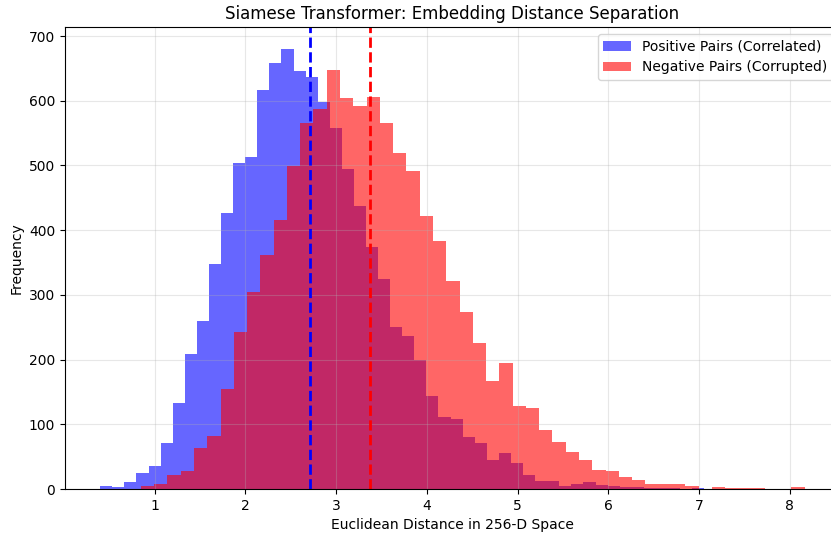


Figure 4: Distribution of embedding distances for correlated and corrupted pairs. The separation between the means is what the Triplet Accuracy measures.

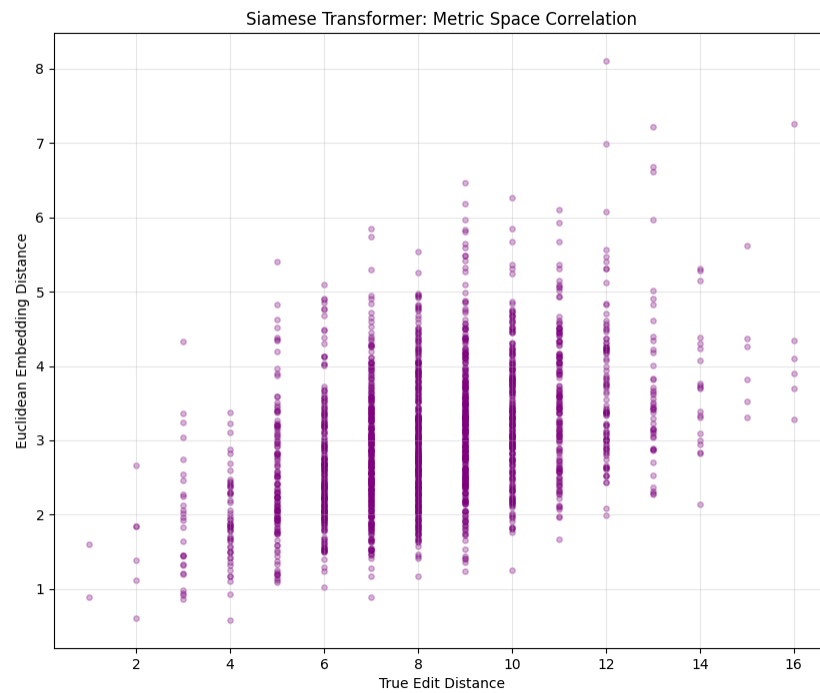


Figure 5: True Levenshtein distance against predicted embedding distance. The spread around the trend line explains the gap between strong absolute error and weaker rank correlation.

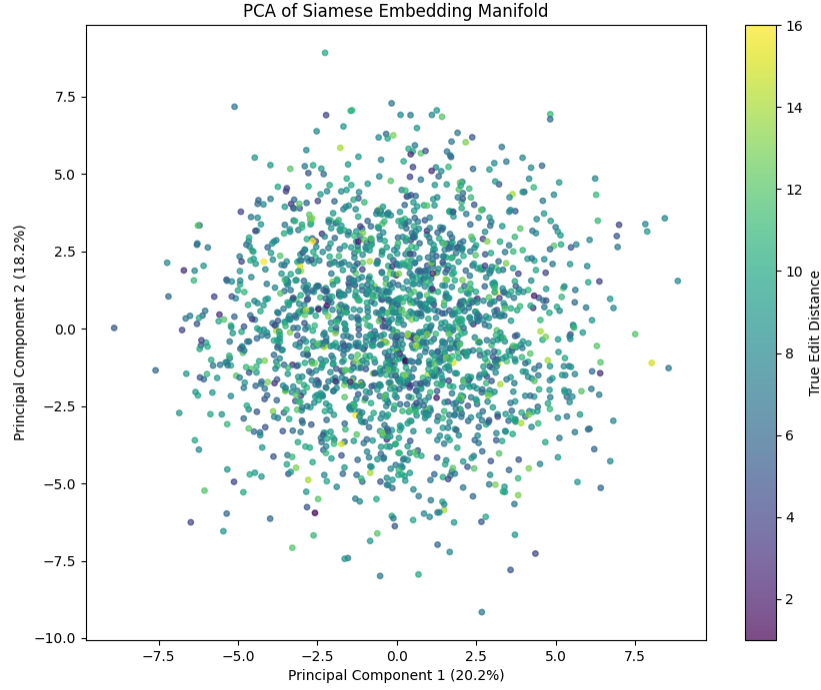


Figure 6: The 256-dimensional embedding projected to two dimensions by Principal Component Analysis, colored by corruption severity. Sequences organize by structural integrity rather than at random.

Discussion and Baseline Comparison

Table 1 places both Transformer phases against classical baselines and a stronger reference architecture from the lab.

Table 1: Comparative Edit Distance Metrics Across Architectures

Model	RMSE ↓	MAE ↓	Pearson ↑	Spearman ↑	Triplet Acc. ↑	Hamming ↓
Transformer (Gen.)	2.482	1.931	0.859	0.856	0.530	0.372
GRU	3.244	2.530	0.592	0.575	0.636	0.310
CNN-ED	2.906	2.263	0.621	0.612	0.684	0.325
CGK	2.622	2.062	0.688	0.693	0.721	N/A
Siamese (Proposed)	2.092	1.670	0.455	0.452	0.7238	N/A
Specialized Lab Arch.	1.541	1.180	0.912	0.914	0.835	0.049

The two Transformer rows tell opposite halves of one story. The generative model has the better correlation of the pair, 0.859 against 0.455, and it is the only learned model in the table that yields a Hamming Loss at all, because it is the only one that prints bits. That advantage is hollow once the Block Error Rate is read alongside it: a decoder that fails 0.9999 of its blocks does not decode. The Siamese model gives up sequence output and, with it, the Hamming column, but it returns the lowest RMSE and MAE of any learned model and the cleanest triplet separation.

The weaker correlation of the Siamese model deserves a plain explanation. Triplet Margin Loss pushes each pair past a fixed margin, which fixes local ordering and tight absolute error, but it never asks the whole scale to line up in rank. So the absolute error drops while Pearson and Spearman trail the convolutional and alignment baselines. The specialized lab architecture remains ahead on every column, and this report does not claim otherwise. What the two phases establish is narrower and still useful: attention captures the geometry of edit operations well, and the metrics a model can report follow directly from whether it generates or embeds.

Future Work

The obvious next step is a model that recovers bit sequences without the shift penalty, which would restore the Bit Error Rate and Hamming Loss that the embedding cannot give. A Connectionist

Temporal Classification head is the natural candidate, since it scores all alignments at once through a blank symbol and tolerates insertions and deletions. A first version did not work. Trained on the full one million sequence set, the loss settled near $\ln 2$ and the model collapsed to predicting the bit marginal, scoring a Bit Error Rate of 0.5859, worse than simply copying the input, which scores 0.0823. A short overfitting test on 128 samples could not drive the error down either, which rules out a plain learning rate problem and points at the architecture. The likely cause is the nearest-neighbor temporal upsampling used to satisfy the spacing requirement of the loss: duplicating each frame produces identical adjacent vectors, which narrows the set of alignments the decoder can express. Replacing that stage with a learned upsampling layer, such as a transposed convolution, is the planned next step.